

Two Seasons, Shifted Dynamics: Decoding Tactical Evolution in IPL 2018 vs 2019

An end-to-end relational database investigation using Google BigQuery to evaluate player performance consistency, team scoring dominance, and bowling containment efficiency.

1,250+ Runs

Multi-Season Dominance

KL Rahul led all batters across 2018–2019 via INNER JOIN aggregations.

6.57 ER

Elite Containment

Rashid Khan claimed 21 wickets while restricting runs under 7.0 per over.

105 Boundaries

Power Hitting Record

Rishabh Pant topped boundary counts through combined arithmetic fields.

ANALYTICAL PROBLEM STATEMENT

Uncovering Hidden Player Efficiency & Team Dominance Behind Raw Totals

Raw match totals hide tactical value. Franchise analysts require refined queries—filtering high scoring volume against strike rate constraints, and evaluating wicket takers against economy rates.



Volume vs Speed Paradox

High scoring volume is ineffective without speed. We query top scorers constrained by `SR ≥ 130`.



Defensive Containment

Isolating bowlers who restrict runs (`ER < 7.0`) versus those taking expensive wickets.



Multi-Season Continuity

Joining 2018 and 2019 seasonal datasets using `INNER JOIN` on `Player` primary keys.

SQL Logic Applied: Conditional `WHERE` filtering, relational joins, and multi-column expressions
Recruiter Skill Takeaway: Translating complex sports business questions into exact analytical SQL parameters.

Portfolio: [Thaarun132](#)

Relational BigQuery Schema Architecture & Query Pipeline

Four seasonal tables hosted under BigQuery database `iplanalysis-508512.IPL_db`.

📊 Batsmen Tables (2018 / 2019)

COLUMN NAME	DATA TYPE	DESCRIPTION
Player	STRING (PK)	Player Identifier Name
Runs	INT64	Total Season Runs Scored
SR	FLOAT64	Batting Strike Rate
Fours / Sixs	INT64	Boundary Counts (4s & 6s)
Fiftys	INT64	Half Centuries Scored



📊 Bowlers Tables (2018 / 2019)

COLUMN NAME	DATA TYPE	DESCRIPTION
Player	STRING (PK)	Player Identifier Name
Wkts	INT64	Total Wickets Taken
ER	FLOAT64	Economy Rate (Runs/Over)
Overs	FLOAT64	Total Overs Bowled
FourWs	INT64	4-Wicket Hauls Count

BigQuery Schema Path: `iplanalysis-508512.IPL_db.*`

Recruiter Skill Takeaway: Relational database hygiene, multi-table schema modeling, and primary key alignment.

Repository: [Thaarun132](#)

TEAM AGGREGATION ANALYSIS

Mumbai Indians Led All Franchises in Aggregate Offensive Output in 2019

Executing a grouped sum aggregation to calculate total runs scored across all batters per team.

```
-- Query 8: Maximum Runs Scored by Team in 2019

SELECT
  Team,
  SUM(Runs) AS TotalTeamRuns
FROM `IPL_db.Batsmen2019`
GROUP BY Team
ORDER BY TotalTeamRuns DESC
LIMIT 1;
```

 BIGQUERY QUERY RESULT

✓ EXECUTED IN 0.22S

ROW	TEAM	TOTALTEAMRUNS
1	Mumbai Indians (MI)	2,430

Insight Summary:

MI's balanced middle order yielded the highest cumulative run output, paving their path to the 2019 IPL title.

SQL Concept Demonstrated: Categorical Aggregation (GROUP BY + SUM())

Recruiter Skill Takeaway: Ability to aggregate row-level player stats into executive team-level KPIs.

Query 8 Verified



BATTING EFFICIENCY SPOTLIGHT

High Volume Plus Elite Speed: Kane Williamson Dominated IPL 2018

Filtering top run scorers maintaining an aggressive average strike rate threshold (SR ≥ 130).

```
-- Query 2: Most Runs in 2018 with SR ≥ 130

SELECT
  Player,
  Runs,
  SR
FROM `IPL_db.Batsman2018`
WHERE SR >= 130
ORDER BY Runs DESC
LIMIT 1;
```

BIGQUERY QUERY RESULT

EXECUTED IN 0.18S

ROW	PLAYER	RUNS	SR
1	Kane Williamson	735	142.44

Key Takeaway:

Williamson captured the Orange Cap by balancing massive scoring volume (735 runs) with high scoring velocity (SR 142.44).

SQL Concept Demonstrated: Composite Conditional Filtering (WHERE + Inequality Comparison)

Recruiter Skill Takeaway: Ability to apply multi-attribute business rules to eliminate low-efficiency stats.

Query 2 Verified

Rashid Khan Combined Heavy Wicket Tallies With Sub-7.0 Run Restraint

Comparing total top wicket taker (Andrew Tye: 24 Wkts) vs defensive bowling under strict economy (ER < 7.0).

```
-- Query 3: Most Wickets in 2018 with ER < 7

SELECT
  Player,
  Wkts,
  ER
FROM `IPL_db.Bowlers2018`
WHERE ER < 7
ORDER BY Wkts DESC
LIMIT 1;
```

BIGQUERY QUERY RESULT

EXECUTED IN 0.19S

ROW	PLAYER	WKTS	ER
1	Rashid Khan	21	6.57

Query 1 Benchmark (Pure Wkts Leader):
Andrew Tye (24 Wkts)

ER: 8.46

SQL Concept Demonstrated: Dual Constraint Evaluation (WHERE ER < 7 + ORDER BY Wkts DESC)

Recruiter Skill Takeaway: Uncovering rare hybrid performers who excel in both offensive and defensive metrics.

Queries 1 & 3 Verified



MULTI-SEASON CONTINUITY

Cross-Season Relational JOINS Reveal Multi-Year Batting Stalwarts

Joining 2018 and 2019 seasonal tables via `INNER JOIN` on `Player` keys.

```
-- Query 4: Most Combined Runs (2018 + 2019)

SELECT
  bat2018.Player,
  (bat2018.Runs + bat2019.Runs) AS TotalRuns
FROM `IPL_db.Batsman2018` AS bat2018
INNER JOIN `IPL_db.Batsmen2019` AS bat2019
  ON bat2018.Player = bat2019.Player
ORDER BY TotalRuns DESC
LIMIT 2;
```

QUERY 4 RESULT: TOP 2 CROSS-SEASON SCORERS

ROW	PLAYER	TOTALRUNS
1	KL Rahul	1,252
2	Shikhar Dhawan	1,018

QUERY 9 RESULT: TOP 50S (COMBINED)

ROW	PLAYER	TOTALFIFTYS
1	KL Rahul / David Warner	12 Fifties

SQL Concept Demonstrated: Multi-Table `INNER JOIN` & Column Addition Expressions

Recruiter Skill Takeaway: Ability to combine disparate annual tables without row duplication or Cartesian products.

Queries 4 & 9 Verified

CALCULATED FIELDS & FILTERING

Derived Fields Highlight Extreme Boundary Impact & Special Hauls

Creating aliased boundary sum calculations (`Fours + Sixs`) and low-over filters.

```
-- Query 5: Max Boundaries (Fours + Sixs) in 2018

SELECT
  Player,
  (Fours + Sixs) AS TotalBoundaries
FROM `IPL_db.Batsman2018`
ORDER BY TotalBoundaries DESC
LIMIT 1;
```

QUERY 5 RESULT: BOUNDARY KING

ROW	PLAYER	TOTALBOUNDARIES
1	Rishabh Pant	105 (68 Fours + 37 Sixes)

Query 6 & 7 Additional Insights:

- Query 6 (4-Wicket Hauls 2019):** Isolated bowlers with FourWs > 0 (e.g. Kagiso Rabada, Lasith Malinga).
- Query 7 (Least Overs Bowled 2018):** Identified part-time bowling options using ORDER BY Overs ASC LIMIT 5.

SQL Concept Demonstrated: Custom Calculated Projections & Boundary Aliasing

Recruiter Skill Takeaway: Creating novel metrics directly inside SQL SELECT projections without modifying underlying tables.

Queries 5, 6, 7 Verified



CANDIDATE TECHNICAL COMPETENCIES

Full-Stack Data Analyst Capabilities: SQL Rigor & Analytics Storytelling

Summary of technical tools and SQL querying techniques showcased throughout this IPL case study.



Advanced Relational SQL

INNER JOINS, multi-table column arithmetic, GROUP BY aggregations, ORDER BY logic, HAVING / WHERE filters.



Google BigQuery

Cloud data warehouse query execution, dataset schema management, and cost-efficient execution planning.



KPI Definition

Formulating new performance indexes combining boundaries, economy rates, strike rates, and total runs.



Data Storytelling

Translating query outputs into executive-level slides formatted for recruiter review and stakeholders.

Candidate Profile: Thaaron132 | Ready for Data Analyst & Business Intelligence roles.

github.com/Thaaron132

DATA ANALYST CANDIDATE

Let's Connect & Build High-Impact SQL Data Solutions

Open for Data Analyst, SQL Developer, and Business Intelligence roles. I bring strong relational querying, cloud data warehousing, and clear executive data storytelling capabilities.



GitHub Repository
github.com/Thaarun132



SQL Solution File
[IPL_Solution.sql](#)